

Biotechnology and Food Engineering

Process Biotechnology

Spring 2026

Tutorial 7 – Oxygen transfer – Solutions

Question 1

The factory you are working in performs batch fermentation of yeast for the purpose of protein production which consists 45% of the yeast. The fermentation goal is to obtain 1000 kg clean and purified protein from each fermentation, so they use a fermenter with the volume of 120 m³.

The yeast doubling time is 72 minute and the growth process is aerobic. The growth yield on oxygen is $0.825 \frac{gX}{gO_2}$ and the critical dissolved oxygen concentration required for growth is 0.125 mM. Air is supplied into the fermenter in pressure of 3 atmospheres, and it is known that the oxygen transfer rate coefficient in the fermenter is 700 hr⁻¹.

Since the oxygen in this fermentation is the limiting factor, in order to reach the required amount of protein, it is necessary to supply air enriched with oxygen. Yet, from economic reasons, the air supplied into the fermenter is being enriched with oxygen gradually, every time the soluble oxygen concentration in the medium drops to the critical point.

The fermentation starts with aeration by normal air (21% oxygen) and the maximal oxygen solubility in this case is 9 ppm. The fermenter is being inoculated with 6 kg yeast, that start its logarithmic growth immediately. As the fermentation progress, the oxygen concentration in the fermenter declines, and when it reaches to the critical concentration, the air is being enriched by 15% (meaning – the oxygen percentage in the new air is 36%). Afterwards, the fermentation keeps on going until the oxygen concentration in the medium decline again, so more 15% oxygen are added to the air. This is how the process goes until the fermentation ends. We can assume that the air flow and its physical and chemical properties do not change due to the oxygen enrichment.

At the end of the fermentation, the fermenter content is transferred to separation processes, and the protein purification yield is 92.6%.

- A. Plan the time schedule of air enrichment with oxygen, from the beginning to the end of the fermentation.**
- B. To avoid wastage, what is the exact oxygen percentage that should be added in the last air enrichment?**

Solution:

Given:

$$Y_{\frac{P}{X}} = 0.45 \frac{gP}{gX}$$

$$W_{P_{clean}} = 1000 \frac{kgP}{ferm.}$$

$$V = 120,000 \text{ lit}$$

$$t_d = 72 \text{ min} = 1.2 \text{ hr}$$

$$Y_{\frac{X}{O_2}} = 0.825 \frac{gX}{gO_2}$$

$$C_{L_{min}} = 0.125 \text{ mM} = 0.125 \frac{\text{mmol}}{\text{lit}} \cdot 32 \frac{g}{\text{mol}} \cdot \frac{\text{mol}}{1000 \text{ mmol}} = 0.004 \frac{gO_2}{\text{lit}}$$

$$P = 3 \text{ atm}$$

$$K_L a = 700 \frac{1}{\text{hr}}$$

$$21\% O_2 \rightarrow C^* = 0.009 \frac{gO_2}{\text{lit}}$$

$$w_{x_0} = 6 \text{ kgX} \rightarrow X_0 = \frac{6000 gX}{120,000 \text{ lit}} = 0.05 \frac{gX}{\text{lit}}$$

$$\eta = 92.6\% \text{ yield} = 0.926$$

A. Time-table for oxygen enrichment.

We start from understand what is happening in the system: the bacteria concentration (X) rising as well as the oxygen uptake rate (OUR). Since $OTR = OUR$, OTR is also rising, thus C_L decline until it reaches to the critical concentration $C_{L_{min}}$. At that stage, we need to enrich

the air with oxygen. To find the required schedule for oxygen enrichment we need to find when C_L reach to its critical concentration, according to:

$$OTR = OUR$$

$$\frac{\mu \cdot X}{Y_{x/o}} = K_L a \cdot (C^* - C_L)$$

By placing $C_L = C_{L_{min}}$ we can find X at the moment that the oxygen reach its critical concentration, and using X we can find the time (t) from the growth equation.

Before we start with time calculations, we first need to know what is the final biomass concentration X_F in this fermentation, according to the amount of product required (1000 kgP):

We can find X_F from the product yield on the biomass:

$$Y_{\frac{P}{X}} = \frac{P_f - P_0}{X_f - X_0} \rightarrow X_f = \frac{P_f}{Y_{\frac{P}{X}}} + X_0$$

We only need to find P_f (all the other parameters are given in the question):

$$W_p = \frac{W_{p_{clean}}}{\eta} = \frac{1000 \text{ kgP}}{0.926} = 1080 \text{ kgP}$$

$$P_f = \frac{1080 \text{ kgP} \cdot \frac{1000 \text{ gP}}{\text{kgP}}}{120,000 \text{ lit}} = 9 \frac{\text{gP}}{\text{lit}}$$

Place all numbers in the equation to find X_f :

$$X_f = \frac{9 \frac{\text{gP}}{\text{lit}}}{0.45 \frac{\text{gP}}{\text{gX}}} + 0.05 \frac{\text{gX}}{\text{lit}} = 20.05 \frac{\text{gX}}{\text{lit}}$$

This is the final biomass concentration of the fermentation.

First enrichment time (Δt_1):

$$\ln\left(\frac{X_{t_1}}{X_0}\right) = \mu \cdot \Delta t_1$$

We know X_0 . We can find the growth rate by:

$$\mu = \frac{\ln 2}{t_d} = \frac{\ln 2}{1.2 \text{ hr}} = 0.578 \frac{1}{\text{hr}}$$

We can find X_{t_1} by:

$$\frac{\mu \cdot X}{Y_{x/o}} = K_L a \cdot (C^* - C_L) \rightarrow X_{t_1} = \frac{K_L a \cdot (C^* - C_L)}{\mu} \cdot Y_{x/o}$$

$$X_{t_1} = \frac{700 \frac{1}{\text{hr}} \cdot (0.009 - 0.004) \frac{gO_2}{\text{lit}}}{0.578 \frac{1}{\text{hr}}} \cdot 0.825 \frac{gX}{gO_2} = 5 \frac{gX}{\text{lit}}$$

Find Δt_1 :

$$\ln\left(\frac{X_{t_1}}{X_0}\right) = \mu \cdot \Delta t_1$$

$$\Delta t_1 = \frac{\ln\left(\frac{5 \frac{gX}{\text{lit}}}{0.05 \frac{gX}{\text{lit}}}\right)}{0.578 \frac{1}{\text{hr}}} = 7.97 \text{ hr}$$

After 7.97 hour, the air is enriched with oxygen to 36%.

Second enrichment time (Δt_2):

We need to pay attention that now C^* is different since the oxygen percentage in the air has changed to 36%. We can find the new C^* from Henry's law:

$$C^* = \frac{P \cdot \%O_2}{H}$$

We can find H by using the initial data:

$$0.009 \frac{gO_2}{\text{lit}} = \frac{3 \text{ atm} \cdot 0.21}{H}$$

$$H = 70 \frac{\text{atm} \cdot \text{lit}}{gO_2}$$

Therefore, while the air is enriched with 36% oxygen, the value of C_2^* is:

$$C_2^* = \frac{3 \text{ atm} \cdot 0.36}{70 \frac{\text{atm} \cdot \text{lit}}{gO_2}} = 0.015 \frac{gO_2}{\text{lit}}$$

Next, we can repeat the same calculations as before to find Δt_2 :

$$X_{t_2} = \frac{700 \frac{1}{\text{hr}} \cdot (0.015 - 0.004) \frac{gO_2}{\text{lit}}}{0.578 \frac{1}{\text{hr}}} \cdot 0.825 \frac{gX}{gO} = 11 \frac{gX}{\text{lit}}$$

$$\Delta t_2 = \frac{\ln\left(\frac{11 \frac{gX}{\text{lit}}}{0.05 \frac{gX}{\text{lit}}}\right)}{0.578 \frac{1}{\text{hr}}} = 9.33 \text{ hr}$$

After 9.33 hour, the air is enriched with oxygen to 51%. We perform the same calculations as before:

$$C_3^* = \frac{3 \text{ atm} \cdot 0.51}{70 \frac{\text{atm} \cdot \text{lit}}{gO_2}} = 0.022 \frac{gO_2}{\text{lit}}$$

$$X_{t_3} = \frac{700 \frac{1}{\text{hr}} \cdot (0.022 - 0.004) \frac{gO_2}{\text{lit}}}{0.578 \frac{1}{\text{hr}}} \cdot 0.825 \frac{gX}{gO} = 18 \frac{gX}{\text{lit}}$$

$$\Delta t_3 = \frac{\ln\left(\frac{18 \frac{gX}{\text{lit}}}{0.05 \frac{gX}{\text{lit}}}\right)}{0.578 \frac{1}{\text{hr}}} = 10.2 \text{ hr}$$

At that point of time the air is enriched to 66%. Again, the same way:

$$C_4^* = \frac{3 \text{ atm} \cdot 0.66}{70 \frac{\text{atm} \cdot \text{lit}}{gO_2}} = 0.028 \frac{gO_2}{\text{lit}}$$

$$X_{t_4} = \frac{700 \frac{1}{hr} \cdot (0.028 - 0.004) \frac{gO_2}{lit}}{0.578 \frac{1}{hr}} \cdot 0.825 \frac{gX}{gO} = 24 \frac{gX}{lit}$$

We see that in this stage the biomass concentration overreach the final required concentration

($X_{t_4} = 24 \frac{gX}{lit} > X_f = 20.05 \frac{gX}{lit}$), therefore we can stop the fermentation.

To conclude, the enrichment time-table is:

#	<i>t</i> (hr)	% <i>O</i> ₂
1	7.97	36
2	9.33	51
3	10.2	66

B. %*O*₂(*t*₃) = ?

At the last enrichment we are required to get to the final biomass concentration of $20.05 \frac{gX}{lit}$.

To do that, we need to find what is the required oxygen percentage in the last addition of enriched air.

The oxygen percentage can be found by Henry's law, to which we need to find the value of oxygen concentration ($C_{4'}^*$) that will lead to the desired final biomass concentration.

Place $X_f = 20.05 \frac{gX}{lit}$ to find $C_{4'}^*$:

$$\frac{\mu \cdot X}{Y_{\frac{x}{o}}} = K_L a \cdot (C^* - C_L)$$

$$C_{4'}^* = \frac{\mu \cdot X}{Y_{\frac{x}{o}} \cdot K_L a} + C_L = \frac{0.578 \frac{1}{hr} \cdot 20.05 \frac{gX}{lit}}{0.825 \frac{gX}{gO_2} \cdot 700 \frac{1}{hr}} + 0.004 \frac{gO_2}{lit} = 0.024 \frac{gO_2}{lit}$$

Use Henry's law to find the oxygen percentage:

$$0.024 \frac{gO_2}{lit} = \frac{3 atm \cdot \%O_2}{70 \frac{atm \cdot lit}{gO_2}}$$

$$\%O_2 = 56\%$$

Question 2

A student's group was visiting a biotechnology factory and was delaying in the control room to hear in detail about the factory's yeast fermentation.

A tired student was accidentally leaning on the oxygen control button, what led to an increase of 2-fold in the oxygen percentage in the air supplied to the fermenter. As a result, the soluble oxygen concentration in the fermenter started to rise up from its original value of 80% of its saturated value.

The student got stressed and tried to lower the oxygen percentages in the air back down so that the soluble oxygen concentration will return to its original values before the accident. But instead of turning the right button, the student turned the speed control button of the stirrer in the fermenter and lowered the stirring speed. The soluble oxygen percentage stopped rising and started declining, and eventually, the system was stabilized on 75% of the saturated value. All the process of change and stabilization was rapid, and at the end, the students left the control room and never looked back...

The original air supply (before the modifications done by the student) was planned so that it will suffice exactly to the maximal oxygen amount required in this process of batch growth from start to end. The fermenter volume is 5 m³ and at the beginning of the fermentation it is being inoculate with 890 gram of yeast. After inoculation, the yeast stays in lag phase for 2 hours and afterwards start logarithmic growth of 15 hours. At the end of the logarithmic growth the fermenter is harvested and transferred to separation processes. It is known that the yeast doubling time in this fermentation is 8603 seconds and that the biomass yield on the oxygen is $0.799 \frac{gX}{gO_2}$.

- A. By what fold will the oxygen transfer coefficient (K_La) in the fermenter rise or drop, due to the changes done by the student?**
- B. As a result of the changes made during the fermentation, and assuming that the limiting factor is the oxygen and not the substrate (relatively to a normal fermentation), after how many hours the fermentation will be finished?**

Solution:

Given:

$$\%O_{2_{new}} = 2\% O_2$$

$$C_L = 0.8C^*$$

$$C_{L_{new}} = 0.75C_{new}^*$$

$$V = 5000 \text{ lit}$$

$$W_{X_0} = 890 \text{ gX} \rightarrow X_0 = \frac{890 \text{ gX}}{5000 \text{ lit}} = 0.178 \frac{\text{gX}}{\text{lit}}$$

$$t_{lag} = 2 \text{ hr}$$

$$t_{log} = 15 \text{ hr}$$

$$t_d = 8603 \text{ sec} = 2.39 \text{ hr} \rightarrow \mu = \frac{\ln 2}{2.39 \text{ hr}} = 0.29 \frac{1}{\text{hr}}$$

$$Y_{\frac{X}{O_2}} = 0.799 \frac{\text{gX}}{\text{gO}_2}$$

A. $\frac{K_La \text{ (new)}}{K_La} = ?$

$$OTR = K_La(C^* - C_L) \rightarrow K_La = \frac{OTR}{(C^* - C_L)}$$

$$\frac{K_L a_{new}}{K_L a} = \frac{\frac{OTR_{new}}{(C^* - C_L)_{new}}}{\frac{OTR}{(C^* - C_L)}}$$

We can assume that $OTR = OUR$ because it was mentioned in the question that the changes happened very quickly, therefore:

$$OUR = \frac{\mu \cdot X}{Y_X \frac{O_2}{O_2}}$$

Since the change is rapid, OUR does not change (the biomass concentration does not change), therefore:

$$OUR_{new} = OUR$$

As well as:

$$OTR_{new} = OTR$$

Hence,

$$\frac{K_L a_{new}}{K_L a} = \frac{\frac{\cancel{OTR_{new}}}{(C^* - C_L)_{new}}}{\frac{\cancel{OTR}}{(C^* - C_L)}} = \frac{(C^* - C_L)}{(C^* - C_L)_{new}}$$

We have the data on C_L , but we should pay attention that due to the change in oxygen percentage also C^* is changing. According to Henry's law:

$$C^* = \frac{P \cdot \%O_2}{H}$$

We can see from Henry's law that C^* is directly correlated to $\%O_2$, so if $\%O_{2_{new}} = 2\% O_2$ then $C^*_{new} = 2C^*$.

$$\frac{K_L a_{new}}{K_L a} = \frac{C^* - 0.8C^*}{2C^* - 0.75C^*_{new}} = \frac{0.2C^*}{2C^* - 0.75 \cdot 2C^*} = \frac{1}{2.5}$$

The new $K_L a$ is 2.5-fold smaller after the changes.

B. $t = ?$

Normally, the fermentation takes 17 hours. But, it is given that after the changes, the limiting factor in the fermentation is the oxygen, which imply that the fermentation is finished when $C_{L\min} = 0$. We need to see how this change affect the fermentation outcome. At the end of the growth we know that:

$$OUR_{max} = \frac{\mu \cdot X_f}{Y_{x/o}} = OTR_{max}$$

$$OTR_{max} = K_L a \cdot (C^* - \cancel{C_{L\min}}) = K_L a \cdot C^*$$

$$\frac{\mu \cdot X_f}{Y_{x/o}} = K_L a \cdot C^*$$

From the equation above we can learn that due to the changes in the values of the oxygen transfer coefficient ($K_L a$) and the oxygen concentration in saturation (C^*), the final biomass concentration (X_f) must change (the growth rate and yield are constant and are not affected by the change).

If X_f changes, it will lead to change in the fermentation period. To find that time period we can use the growth equation:

$$\ln \frac{X_{f_{new}}}{X_0} = \mu \cdot t_{log_{new}}$$

We have the growth rate and X_0 given in the question. We need to find $X_{f_{new}}$:

$$\frac{\mu \cdot X_{f_{new}}}{Y_{x/o}} = (K_L a \cdot C^*)_{new} \rightarrow X_{f_{new}} = \frac{Y_{x/o} \cdot (K_L a \cdot C^*)_{new}}{\mu}$$

We don't have the values for $K_L a \cdot C^*$, but we can use our findings from section A - the ratio between $\frac{K_L a_{new}}{K_L a} = \frac{1}{2.5}$ and $C^*_{new} = 2C^*$:

$$\frac{X_{f_{new}}}{X_f} = \frac{\frac{Y_{x/o} \cdot (K_L a \cdot C^*)_{new}}{\mu}}{\frac{Y_{x/o} \cdot (K_L a \cdot C^*)}{\mu}} = \frac{K_L a_{new}}{K_L a} \cdot \frac{C^*_{new}}{C^*} = \frac{1}{2.5} \cdot 2 = 0.8$$

$$X_{f_{new}} = 0.8 \cdot X_f$$

To find X_f we will use the growth equation with the information we have about the fermentation before the change:

$$\ln\left(\frac{X_f}{X_0}\right) = \mu \cdot t_{log}$$

$$\ln\left(\frac{X_f}{0.178 \frac{gX}{lit}}\right) = 0.29 \frac{1}{hr} \cdot 15 \text{ hr}$$

$$X_f = 13.79 \frac{gX}{lit}$$

Now we can find $X_{f_{new}}$:

$$X_{f_{new}} = 0.8 \cdot 13.79 \frac{gX}{lit} = 11 \frac{gX}{lit}$$

We place it in the growth equation to find $t_{log_{new}}$:

$$t_{log_{new}} = \frac{\ln\left(\frac{X_{f_{new}}}{X_0}\right)}{\mu} = \frac{\ln\left(\frac{11 \frac{gX}{lit}}{0.178 \frac{gX}{lit}}\right)}{0.29 \frac{1}{hr}} = 14.23 \text{ hr}$$

Therefore, the total time of fermentation after the changes:

$$t_{tot} = 2 \text{ hr} + 14.23 \text{ hr} = 16.23 \text{ hr}$$

$$\mathbf{t_{tot} = 16.23 \text{ hr}}$$

Question 3

Bacteria grow in continuous fermenter. The dilution rate in the fermenter is 0.8 per hour, the glucose concentration in the fresh feed is $9 \frac{g}{lit}$ and 99% of it is utilized by the bacteria in the fermenter. The bacteria yield on glucose is 45% and their yield on oxygen is $1.2 \frac{gX}{gO_2}$. Monod constant is $90 \frac{mg}{lit}$. The amount of heat released in this fermentation is $3,648 \frac{Kcal}{day}$. Air is being supplied in pressure of 1.75 atmospheres and temperature of $35^{\circ}C$. The maximal oxygen solubility in the fermentation temperature is 8 ppm, and in steady state the soluble oxygen concentration in the fermenter is 15% of the saturated soluble oxygen concentration.

- A. What is the biomass production rate ($\frac{g}{hr}$)?**
- B. What is the required air flow rate ($\frac{lit}{hr}$), under the assumption that only 8% of the air are dissolving?**
- C. You initiated the integration of a new air dispenser (Sparger) in the fermenter that will enable higher $K_L a$ value in the system, 2-fold than what was before. What will be the soluble oxygen concentration in the fermenter after the system will stabilize again?**
- D. In your opinion, what is the maximal possible increase in the dilution rate without causing rinsing?**

Solution:

Given:

$$D = 0.8 \frac{1}{hr}$$

$$S_0 = 9 \frac{gS}{lit}$$

$$\Delta S = 0.99 \cdot 9 \frac{gS}{lit} \rightarrow S_f = 0.09 \frac{gS}{lit}$$

$$Y_{\frac{X}{S}} = 0.45 \frac{gX}{gS}$$

$$Y_{\frac{x}{o}} = 1.2 \frac{gX}{gO_2}$$

$$K_s = 0.09 \frac{g}{lit}$$

$$Q_f = 3,648 \frac{kcal}{day} = 152 \frac{kcal}{hr}$$

$$P = 1.75 atm$$

$$T = 35^\circ C = 308^\circ K$$

$$C^* = 0.008 \frac{gO_2}{lit}$$

$$C_L = 0.15 \cdot C^*$$

A. $G_x = ?$

To find the production rate we need to use $G_x = F \cdot X_f$.

The question deals with a standard continuous fermentation. We don't have neither F or V (to use the information given on D).

We can use Cooney equation:

$$Q_{ferm} = 3.74 \frac{kcal}{gO_2} \cdot OUR \cdot V$$

In a continuous fermentation, OUR is constant and equals to:

$$OUR = \frac{\mu \cdot X_f}{Y_{x/o}}$$

Moreover, from biomass balance under the standard assumptions we know that:

$$\mu = D = \frac{F}{V}$$

Place both expressions in Cooney equation:

$$Q_{ferm} = 3.74 \frac{kcal}{gO_2} \cdot \frac{F}{V} \cdot \frac{X_f}{Y_{\frac{x}{o}}} \cdot \cancel{V}$$

Replace F with $\frac{G_x}{X_f}$ to get:

$$Q_{ferm} = 3.74 \frac{kcal}{gO_2} \cdot \frac{G_x}{Y_{x/o}}$$

$$G_x = \frac{Q_{ferm} \cdot Y_{x/o}}{3.74 \frac{kcal}{gO_2}} = \frac{152 \frac{kcal}{hr} \cdot 1.2 \frac{gX}{gO_2}}{3.74 \frac{kcal}{gO_2}}$$

$$G_x = 48.77 \frac{gX}{hr}$$

B. L.P.H = ? (8% dissolve)

We can find the oxygen uptake rate (*OUR*) and from it we can convert to air flow rate:

$$OUR = \frac{\mu \cdot X_f}{Y_{x/o}} = \frac{\frac{F}{V} \cdot X_f}{Y_{x/o}} = \frac{G_x}{V \cdot Y_{x/o}}$$

We don't have the fermenter volume so we will continue with the expression of $OUR \cdot V$:

$$OUR \cdot V = \frac{G_x}{Y_{x/o}} = \frac{48.77 \frac{gX}{hr}}{1.2 \frac{gX}{gO_2}} = 40.64 \frac{gO_2}{hr}$$

Now we only need to convert between oxygen flow rate to air flow rate:

$$L.P.H = 40.64 \frac{gO_2}{hr} \cdot \frac{1 \text{ mol}_{O_2}}{32 \text{ g}_{O_2}} \cdot \frac{0.082 \frac{lit_{O_2} \cdot atm}{K \cdot mol_{O_2}} \cdot 308 K}{1.75 atm} \cdot \frac{100\% lit_{air}}{21\% lit_{O_2}} \cdot \frac{1}{0.08}$$

$$L.P.H = 1,091 \frac{lit_{air}}{hr}$$

C. $C_{L_{new}} = ?$

$$\text{Given: } K_L a_{new} = 2 \cdot K_L a$$

OUR does not change due to the modification done in the fermenter, so *OTR* will also maintain:

$$OTR_{new} = OTR$$

$$K_L a_{new}(C^* - C_{L_{new}}) = K_L a(C^* - C_L)$$

Place $K_L a_{new} = 2 \cdot K_L a$:

$$2 \cdot K_L a(C^* - C_{L_{new}}) = K_L a(C^* - C_L)$$

Divide in $K_L a \cdot C^*$:

$$2 \cdot \left(1 - \frac{C_{L_{new}}}{C^*}\right) = \left(1 - \frac{C_L}{C^*}\right)$$

Given in the question that $C_L = 0.15C^*$, therefore:

$$2 \cdot \left(1 - \frac{C_{L_{new}}}{C^*}\right) = \left(1 - \frac{0.15C^*}{C^*}\right)$$

$$\frac{C_{L_{new}}}{C^*} = 0.575$$

$$C_{L_{new}} = 0.575 \cdot C^* = 0.5757 \cdot 0.008 \frac{gO_2}{lit} = 0.0046 \frac{gO_2}{lit}$$

$$C_{L_{new}} = 4.6 \text{ ppm}$$

D. $D_C = ?$

We know that:

$$D_C = \mu_{max}$$

Therefore, we need to find μ_{max} by using Monod equation:

$$\mu = \mu_{max} \cdot \left(\frac{S_f}{K_s + S_f} \right)$$

Place all parameters:

$$0.8 \frac{1}{hr} = \mu_{max} \cdot \left(\frac{0.09 \frac{gS}{lit}}{0.09 \frac{g}{lit} + 0.09 \frac{gS}{lit}} \right)$$

$$D_C = \mu_{max} = 1.6 \text{ hr}^{-1}$$

Meaning, we can increase the dilution rate by 2.